Assignment 4

Inspecting HTTP packets using Wireshark Packet Sniffer

Deadlines: Batch1: 27/7/26 Batch2: 29/7/26

Installation

To install Wireshark on Ubuntu, follow these steps.

Method 1: Install from Ubuntu Repository (Recommended)

Step 1: Update the package list

$ sudo apt update

Step 2: Install Wireshark

$ sudo apt install wireshark -y

During installation, you may see the prompt:

Should non-superusers be able to capture packets?

Choose:

Yes

This allows users in the wireshark group to capture packets without running Wireshark as root.

Step 3: Add your user to the Wireshark group

sudo usermod -aG wireshark $USER

Step 4: Apply the new group membership

Either log out and log back in, or run:

newgrp wireshark

Step 5: Verify installation

wireshark --version

Example output:

Wireshark 4.x.x

Step 6: Launch Wireshark

From the terminal:

wireshark

Or search for Wireshark in the Ubuntu Applications menu.

Generate DNS Traffic

Open another terminal and run:

nslookup google.com

or

dig google.com

If dig is not installed:

sudo apt install dnsutils

Then run:

dig google.com

You should now see the DNS query and response packets appear in Wireshark.

Useful Wireshark Filters

Purpose	**Filter**
DNS packets	dns
UDP only	udp
DNS over UDP	udp.port == 53
TCP only	tcp
ICMP packets	icmp
HTTP	http
HTTPS (TLS)	tls

Purpose	Filter
ARP	arp
IPv4	ip
IPv6	ipv6

Troubleshooting

If you get "Couldn't run /usr/bin/dumpcap in child process: Permission denied", ensure your user is in the wireshark group:

groups

If wireshark is not listed, run:

sudo usermod -aG wireshark $USER

newgrp wireshark

Then restart your session.

Assignment

1. The Basic HTTP GET/response interaction

Explore HTTP by downloading a very simple HTML file - one that

is very short, and contains no embedded objects. Do the following:

1. Start up your web browser.

2. Start up the Wireshark packet sniffer, as described in the Introductory lab (but

don’t yet begin packet capture). Enter “http” (just the letters, not the quotation

marks) in the display-filter-specification window, so that only captured HTTP

messages will be displayed later in the packet-listing window.

3. Wait a bit more than one minute (we’ll see why shortly), and then begin

Wireshark packet capture.

4. Enter the following to your browser

http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file1.html

Your browser should display the very simple, one-line HTML file.

5. Stop Wireshark packet capture.

By looking at the information in the HTTP GET and response messages, answer the following questions.

1. Is your browser running HTTP version 1.0 or 1.1? What version of HTTP is the server running?
2. What languages (if any) does your browser indicate that it can accept to the server?
3. What is the IP address of your computer? Of the gaia.cs.umass.edu server?
4. What is the status code returned from the server to your browser?
5. When was the HTML file that you are retrieving last modified at the server?
6. How many bytes of content are being returned to your browser?
7. By inspecting the raw data in the packet content window, do you see any headers within the data that are not displayed in the packet-listing window? If so, name one.

2. The HTTP CONDITIONAL GET/response interaction

Most web browsers perform object caching and thus perform a conditional GET when retrieving an HTTP object. Before performing the steps below, make sure your browser's cache is empty.

Now do the following:

• Start up your web browser, and make sure your browser's cache is cleared, as discussed above.

• Start up the Wireshark packet sniffer

• Enter the following URL into your browser

http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file2.html

Your browser should display a very simple five-line HTML file.

• Quickly enter the same URL into your browser again (or simply select the refresh button on your browser)

• Stop Wireshark packet capture, and enter "http" in the display-filter-specification

window, so that only captured HTTP messages will be displayed later in the packet-listing window.

• (Note: If you are unable to run Wireshark on a live network connection, you can use the http-ethereal-trace-2 packet trace to answer the questions below; see footnote 1. This trace file was gathered while performing the steps above on one of the author's computers.)

Answer the following questions:

8. Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an "IF-MODIFIED-SINCE" line in the HTTP GET?

9. Inspect the contents of the server response. Did the server explicitly return the contents of the file? How can you tell?

10. Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an "IF-MODIFIED-SINCE:" line in the HTTP GET? If so, what information follows the "IF-MODIFIED-SINCE:" header?

11. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.

3. Retrieving Long Documents

The documents retrieved have been simple and short HTML files. Let's next see what happens when we download a long HTML file. Do the following:

• Start up your web browser, and make sure your browser's cache is cleared, as discussed above.

• Start up the Wireshark packet sniffer

• Enter the following URL into your browser

http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file3.html

Your browser should display the rather lengthy US Bill of Rights.

• Stop Wireshark packet capture, and enter "http" in the display-filter-specification window, so that only captured HTTP messages will be displayed.

Answer the following questions:

12. How many HTTP GET request messages did your browser send? Which packet number in the trace contains the GET message for the Bill or Rights?

13. Which packet number in the trace contains the status code and phrase associated with the response to the HTTP GET request?

14. What is the status code and phrase in the response?

15. How many data-containing TCP segments were needed to carry the single HTTP response and the text of the Bill of Rights?

4. HTML Documents with Embedded Objects

Inspect what happens when your browser downloads a file with embedded objects, i.e., a file that includes other objects (in the example below, image files) that are stored on another server(s).

Do the following:

• Start up your web browser, and make sure your browser's cache is cleared, as discussed above.

• Start up the Wireshark packet sniffer

• Enter the following URL into your browser

http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file4.html

Your browser should display a short HTML file with two images. These two images are referenced in the base HTML file. That is, the images themselves are not contained in the HTML; instead the URLs for the images are contained in the downloaded HTML file. As discussed in the textbook, your browser will have to retrieve these logos from the indicated web sites. Our publisher's logo is retrieved from the gaia.cs.umass.edu web site. The image of the cover for our 5th edition (one of our favorite covers) is stored at the caite.cs.umass.edu server. (These are two different web servers inside cs.umass.edu).

• Stop Wireshark packet capture, and enter "http" in the display-filter-specification window, so that only captured HTTP messages will be displayed.

Answer the following questions:

16. How many HTTP GET request messages did your browser send? To which

Internet addresses were these GET requests sent?

17. Can you tell whether your browser downloaded the two images serially, or whether they were downloaded from the two web sites in parallel? Explain.